

MetricMind

Month 1 Project Report

Governed Conversational Business Intelligence Prototype

Prepared for Advanced Data Analytics Portfolio	Project status Month 1 MVP completed and tested
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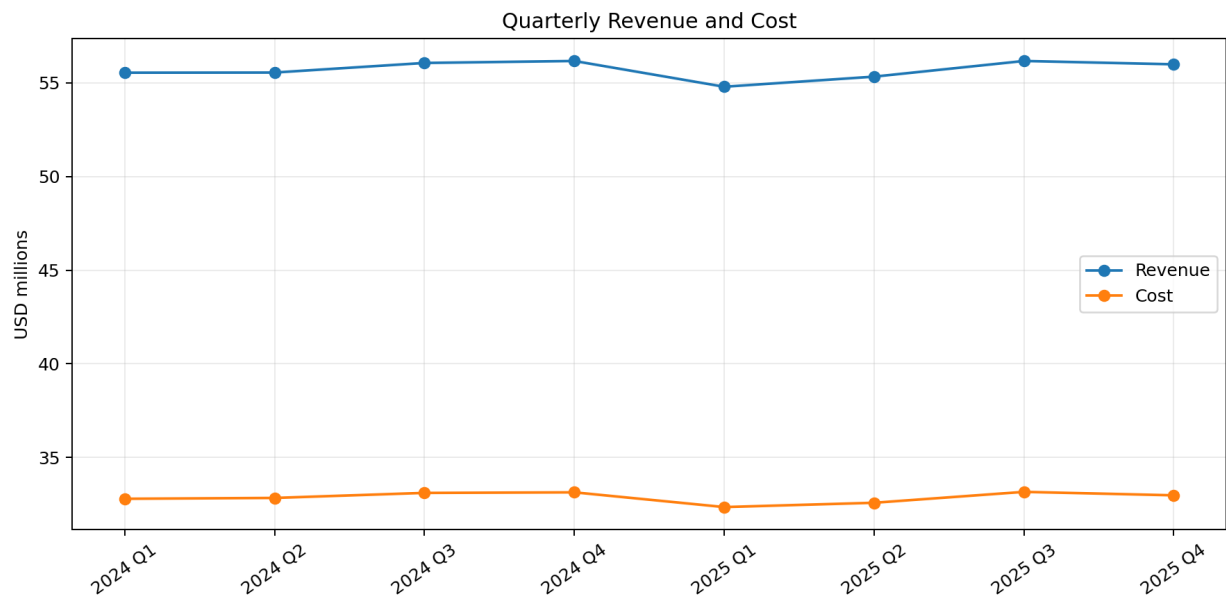
Date: 18 August 2026

1. Executive Summary

MetricMind is a governed conversational BI prototype designed to answer business questions using approved semantic definitions instead of unrestricted text-to-SQL. The Month 1 implementation uses synthetic corporate sales data, a YAML semantic layer, a deterministic intent parser, a semantic query engine, an audit log, and a Streamlit interface.

The project demonstrates the core principle that business metrics such as revenue, cost, profit, and margin should be defined once and reused consistently. The same engine powers summary KPIs, regional or country breakdowns, and a first root-cause analysis workflow.

Dataset rows	24,123
Revenue	\$445,617,932
Cost	\$262,964,387
Profit	\$182,653,545
Overall margin	41.0%



2. Business Problem

Uncontrolled text-to-SQL systems can invent joins, misunderstand time periods, apply inconsistent formulas, and produce numbers that differ from official reports. This creates distrust among finance, sales, and operations teams.

MetricMind addresses this problem by placing a semantic layer between the user and the data. The parser identifies an approved metric and filters, the query compiler produces a structured payload, and the engine calculates the result using governed definitions.

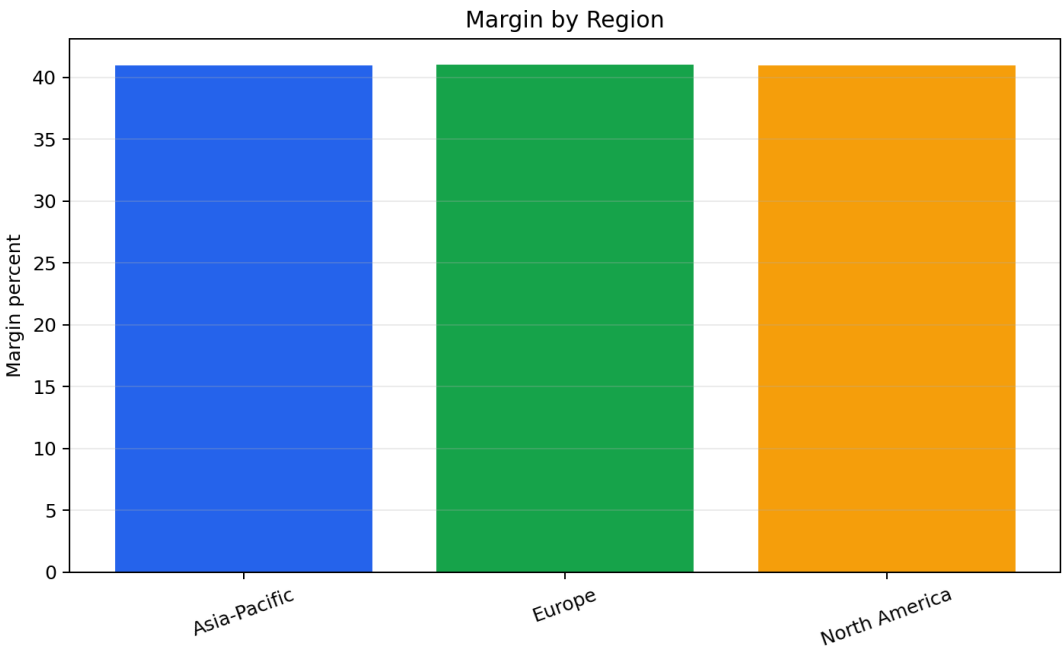
Primary Month 1 use cases

Use case	Example	Output
Metric lookup	Show Q3 2025 revenue	Governed KPI
Breakdown	Show European sales by country	Bar chart and table
Root cause	Why did European margins drop last quarter?	Cost-driver analysis
Transparency	View semantic API payload	Auditable JSON payload

3. Technical Architecture

The Month 1 architecture is deliberately lightweight and reproducible. It can run locally with Python and Streamlit and does not require cloud credentials or an LLM API key.

Layer	Implementation	Purpose
Data	Synthetic CSV dataset	Provides repeatable demo data
Semantic layer	semantic_layer/metrics.yml	Defines approved metrics and dimensions
Orchestration	Deterministic parser	Maps natural language to structured intent
Query engine	Python and pandas	Executes governed calculations
Interface	Streamlit and Plotly	Displays KPIs, tables, and charts
Governance	Audit JSONL log and tests	Supports traceability and repeatability



4. Month 1 Development Plan and Completion

Week	Planned work	Completion evidence
Week 1	Generate mock corporate sales data and define metrics	24 data rows; revenue, cost, profit, and margin definitions
Week 2	Build parser and semantic query compiler	Natural-language queries converted into governed payloads
Week 3	Add dashboard, charts, and root-cause analysis	Streamlit interface with Plotly charts and cost-driver breakdown
Week 4	Testing, documentation, and packaging	3 automated tests passed; README and GitHub-ready structure

5. Results and Validation

The dataset contains two years of daily synthetic sales records across Europe, North America, and Asia-Pacific. The generated data is deterministic because the random generator uses a fixed seed. This makes test results and demonstrations repeatable.

The automated test suite validates three critical behaviors. First, identical governed questions return identical values. Second, European breakdown questions apply the correct region filter and country dimension. Third, margin values remain within the valid 0 to 1 range.

Year	Revenue	Cost	Margin
2024	\$223,330,093	\$131,884,550	40.9%
2025	\$222,287,839	\$131,079,837	41.0%

6. Limitations and Month 2 Roadmap

This is a portfolio-quality Month 1 prototype rather than a production deployment. The data is synthetic, the parser is deterministic, and the semantic engine runs locally. It does not yet include authentication, a cloud warehouse, streaming data, or a real LLM.

Month 2 should introduce a formal semantic API such as Cube or dbt Semantic Layer, LLM function calling with schema-constrained outputs, a FastAPI service, a database-backed audit store, role-based access control, and deployment using Docker. A governance test suite should also compare model answers against official metric fixtures.

7. Conclusion

MetricMind establishes a strong first-month foundation for trustworthy AI-assisted analytics. The central deliverable is not only a chat interface; it is the governed contract between business language, metric definitions, query execution, and auditability.

Repository contents include source code, generated data, semantic definitions, tests, README documentation, and this report.